Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.23	20.25	
worker nodes in service, mean	6.00	6.00	-0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.02	2.02	+0.1%
power (W), mean	630	641	+1.7%
power (W), peak	651	678	+4.1%
energy (Wh)	212	216	+1.8%
energy, parked workers still on at idle power (Wh)	212	216	+1.8%
CPU used (cores), mean	0.911	1.21	+32.3%
CPU allocatable (cores), mean	24.00	24.00	-0.0%
utilisation (used / allocatable)	0.038	0.050	+32.3%
energy per core-hour (Wh)	691	531	-23.1%
node-hours per core-hour	6.59	4.98	-24.4%
pending pods, pod-minutes	0.000	0.000	0
pending pods, peak	0.000	0.000	0
HPA replicas, mean	9.31	9.52	+2.3%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.533	0.783	+46.9%
requests timed	1,040	1,105	+6.2%
time over the response line (% of samples)	0.769	0.000	-100.0%
response time (ms), mean	158	90.26	-43.0%
response time (ms), median	132	82.20	-37.7%
response time (ms), 95th percentile	315	124	-60.5%
response time (ms), 99th percentile	484	149	-69.3%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.1266	5.6845	-20.2%	[-3.0190, +0.1703]	STEADY, no significant difference
kwh_per_core_hour	0.7448	0.6018	-19.2%	[-0.2961, +0.0187]	STEADY, no significant difference
utilisation	0.0376	0.0494	+31.5%	[-0.0002, +0.0251]	STEADY, no significant difference
pending_pod_minutes_per_hour	0.0000	0.0000	n/a	[+0.0000, +0.0000]	STEADY, no significant difference
hpa_shortfall_minutes_per_hour	1.5299	2.2261	+45.5%	[-2.3457, +4.4522]	STEADY, no significant difference

What Omni-Compass did

• decisions: 120 (one per minute)

• HPA target writes: 0

• node-pool resizes: 3

• scheduling-floor adds (pending pods): 0

• power-cap pod resizes (in place): 0

• rollout actions: 4

• power cap decided per minute: 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 0.695 0.695 0.695 1.0 0.695 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	617	620	7.00	6.00	0.000	0.000	0.561	0.661
1	6.00	6.00	620	627	7.00	7.00	0.000	0.000	0.656	0.802
2	6.00	6.00	616	625	7.00	8.50	0.000	0.000	0.559	0.774
3	6.00	6.00	625	638	7.00	9.25	0.000	0.000	0.771	1.12
4	6.00	6.00	636	645	9.00	10.00	0.000	0.000	1.10	1.31
5	6.00	6.00	634	646	9.67	10.00	0.000	0.000	1.01	1.35
6	6.00	6.00	635	645	10.00	10.00	0.000	0.000	1.05	1.40
7	6.00	6.00	646	673	10.00	10.00	0.000	0.000	1.33	2.05
8	6.00	6.00	649	670	10.00	10.00	0.000	0.000	1.40	1.99
9	6.00	6.00	645	670	10.00	10.00	0.000	0.000	1.34	2.00
10	6.00	6.00	645	658	10.00	10.00	0.000	0.000	1.32	1.69
11	6.00	6.00	618	624	10.00	10.00	0.000	0.000	0.627	0.745
12	6.00	6.00	620	623	10.00	10.00	0.000	0.000	0.627	0.746
13	6.00	6.00	623	632	10.00	10.00	0.000	0.000	0.740	0.991
14	6.00	6.00	634	651	10.00	10.00	0.000	0.000	1.04	1.47
15	6.00	6.00	635	651	10.00	10.00	0.000	0.000	1.05	1.48
16	6.00	6.00	633	648	10.00	10.00	0.000	0.000	1.01	1.39
17	6.00	6.00	628	632	10.00	10.00	0.000	0.000	0.897	0.976
18	6.00	6.00	617	621	10.00	10.00	0.000	0.000	0.562	0.677
19	6.00	6.00	621	624	10.00	10.00	0.000	0.000	0.705	0.728
20		6.00		624		10.00		0.000		0.746

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.